W-C-1422/1B
November 8, 1972
SUPERSEDING
Int. Fed. Spec. W-C-001422/1A
January 25, 1971

FEDERAL SPECIFICATION

CAMERA, TELEVISION, MONOCHROME, TYPE I (CLOSED CIRCUIT, GENERAL PURPOSE) (DETAIL SPECIFICATION)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

The complete requirements for procuring type I television cameras described herein shall consist of this document and the issue in effect of Federal Specification W-C-1422.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the detailed requirements for a general purpose monochrome closed circuit television camera.

1.2 Classification.

1.2.1 Style, and mode of synchronization and interlace. Television cameras furnished under this specification shall be of the following style, and mode of synchronization and interlace, as specified (see 6.2):

Style:

Style A - Local operating control.
Style B - Remote operating control.

Modes of synchronization and interlace:

Mode 1 - Positive 2:1 interlace, synchronized from an internal synchronizing generator.
Mode 2 - Positive 2:1 interlace, synchronized from an external synchronizing generator.
Mode 3 - Random interlace, synchronized from an internal synchronizing generator.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Electronic Industries Association (EIA) Standards:

RS-170 - Electrical Performance Standards Monochrome Television Studio Facilities.
RS-330 - Electrical Performance Standards for Closed Circuit Television Cameras 525/60 Interlaced 2:1.
EIA Resolution Chart - 1956 With Gray Scale Overlay Strips.

(Copies may be obtained from the Electronic Industries Association, Engineering Department, 2001 Eye Street, N. W., Washington, D. C. 20006.)

3. REQUIREMENTS

3.1 General. The type I camera shall consist of a single self-contained unit that incorporates the camera tube plus all the electronic circuitry and components necessary to comply with the requirements as specified herein and in W-C-1422. The camera shall be compatible with closed circuit television systems of any size from a simple two unit system consisting of a camera and viewing monitor, to a complex arrangement of cameras, switching equipment, sync generator, viewing monitors, and tape recording facilities.

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3.2 Design and construction. The design of the camera shall be such that it can be supplied as either a style A or style B camera (see 1.2.1). It shall be possible to change the style at any time to accommodate a change in requirements.

3.3 Scanning.

3.3.1 Horizontal scanning. Horizontal scanning shall be 525 lines per frame. Horizontal scanning, when synchronized from internal circuitry, shall be crystal or automatic frequency controlled.

3.3.2 Vertical scanning. The vertical scanning rate shall be 60 fields, 30 frames per second.

3.4 Geometric distortion. Geometric distortion shall not exceed 3 percent of picture height.

3.5 Synchronization and interlace. The camera shall be capable of operation in any one of the modes of synchronization and interlace as listed herein (see 1.2.1, modes of synchronization and interlace) and shall be designed so that it will be possible to change the mode of synchronization and interlace at any time to accommodate a change in requirements. Changing from one mode of synchronization and interlace to another shall be accomplished by a simple procedure, such as a switch operation or plug-in module. It shall not be necessary to disassemble the camera or to solder or unsolder any wires or connections to effect the change. When delivered, the camera shall operate in the mode of synchronization and interlace as specified in the procurement document (see 6.2).

3.5.1 External synchronization. When operated mode 2, the camera shall accept synchronizing signals consisting of blanking, horizontal drive, and vertical drive pulses that conform to EIA RS-330 and EIA RS-170. The camera shall operate in accordance with the requirements, as specified, from input synchronizing signal levels of 4.0 volts $\pm$ 0.5 volts, peak-to-peak, negative going.

3.6 Resolution.

3.6.1 Horizontal limiting resolution. With an EIA Resolution Chart with Gray Scale overlay strips uniformly illuminated to project not more than one (1) foot candle illumination on the face plate of the camera tube and the camera operating positive 2:1 interlace from an internal sync generator and using a 25 mm f/1.4 lens, set at the f/1.4 stop, the horizontal limiting resolution shall equal or exceed 600 lines center, 400 lines each corner.

3.6.2 Horizontal resolution response. When measured with the camera operating in accordance with the requirements of 3.6.1, the minimum horizontal resolution response shall be as follows:

TV Lines	Resolution Response In Percent
100	95
200	75
300	50
400	30
500	20
600	10
700	5
800	2

3.6.3 Vertical limiting resolution. The vertical limiting resolution, when measured with the camera operating in accordance with 3.6.1 shall equal or exceed 325 lines center, 300 lines each corner.

3.7 Video output. In addition to any other outputs that may be provided, the camera shall have at least one, 75 ohm source terminated composite video output. The composite video output signal level shall be adjustable to meet the signal level requirements of EIA RS-170 and EIA RS-330.

3.8 Gray scale. When operated in accordance with the requirements of 3.6.1, the camera shall resolve not less than ten shades of gray (see EIA Resolution Chart Gray Scale Overlay Strips).

3.9 Synchronizing signal waveform. The synchronizing waveform of the composite video signal, measured at the output of the camera, shall conform to the following synchronizing signal requirements, as specified for the different modes of synchronization and interlace.

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3.9.1 Mode 1 operation. When the camera is operated positive 2:1 interlace, synchronized from an internal synchronizing generator, the synchronizing signal waveform shall meet the dimensions and tolerances given in EIA RS-170 or EIA RS-330, as applicable (see 6.2).

3.9.2 Mode 2 operation. When the camera is operated positive 2:1 interlace, synchronized from an external synchronizing generator, in accordance with 3.5.1, the synchronizing signal waveform shall meet the dimensions and tolerances given in EIA RS-170 or EIA RS-330, as applicable (see 6.2).

3.9.3 Mode 3 operation. When the camera is operated random interlace, from an internal synchronizing generator, the synchronizing signal waveform for the horizontal blanking width, horizontal sync pulse, horizontal front porch, horizontal back porch, vertical sync pulse, and vertical blanking shall meet the dimensions and tolerances for these parameters as given in EIA RS-330.

3.10 Video amplifier.

3.10.1 Amplitude frequency response. The amplitude frequency response of the video amplifier, as measured from the input of the camera preamplifier to the video output terminal, shall be ± 2 dB from 60 Hz to 8 mHz.

3.10.2 Low-frequency square-wave response. The tilt imparted to a low-frequency square-wave test signal, whose width measures between 40 to 50 percent of one field time, shall not exceed 5 percent of the peak-to-peak square-wave amplitude.

3.10.3 High frequency transient response. The amount of overshoot or ringing caused by a step function signal having a rise time that measures less than 50 nanoseconds, shall not exceed 5 percent of the peak-to-peak amplitude of the transition.

3.11 Signal to noise ratio. When the camera is operated in accordance with the requirements of 3.6.1, the ratio of the peak-to-peak white signal to the RMS noise shall be greater than 35 dB.

3.12 Automatic light range. The camera shall have an automatic light compensation circuit that will compensate for not less than a 4000 to 1 change in average scene illumination with less than a 2 to 1 change in peak-to-peak output video level. The automatic control shall cover the normal target voltage operating range of the camera tube.

3.13 The camera shall provide specified performance over a camera cable up to 1000 feet long.

3.14 Camera tube. The camera tube shall be EIA registered. The camera tube shall conform to the EIA registration ratings, the manufacturers' published characteristics, and such additional requirements as specified herein. The camera tube shall be equivalent to a one (1) inch vidicon that employs a high sensitivity photoconductor that has low-lag characteristics at low light level operation and has a spectral sensitivety characteristic that is maximum at approximately 5500 angstroms. The camera tube shall provide a picture that is free of noticeable impairments such as spots, blemishes, smudge lines, streaks, or a background which is mottled, grainy or uneven. Flutter and beam splitting shall not be exhibited and the tube shall not be abnormally microphonic.

3.14.1 Shading. Shading shall not exceed 20 percent from center to edge of the picture.

3.14.2 Persistence. The camera tube shall have a persistence of less than 2 percent image retention 2 seconds after removing illumination under conditions of 0.02 microampere dark current, 0.1 microampere peak signal current and illumination for 15 seconds.

3.15 Controls.

3.15.1 Operating controls. It shall be possible to operate all primary operating controls from a remote control unit or they may be operated locally without the remote control unit. The style A camera shall have primary operating controls consisting of not less than a Power On/Off Switch, Beam, Target, Electrical Focus, and Optical Focus. Operating controls on the camera may be recessed screwdriver adjustments. The style B camera shall be a style A camera with a remote control unit and connecting cable as an integral part of the camera. The remote control unit shall have primary operating controls consisting of not less than a Power On/Off Switch, Beam, Target, Electrical Focus. Remote controls shall be mounted on a panel capable of being mounted in a 19-inch rack. The remote control unit shall be capable of controlling the camera up to a distance of 1000 feet from the camera.

3.16 Environment. When specified (see W-T-1422, 6.1), the camera shall show no degradation in performance below that specified herein and in W-C-1422 when subjected to any of the following conditions.

3.16.1 Humidity. Operation at relative humidity up to 95 percent.

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3.16.2 Temperature. Continuous operation over the range of 32°F to 130°F.

3.16.3 Altitude. Operation at altitude up to 10,000 feet.

3.16.4 Vibration. Operation after vibration varying uniformly over a 5 to 35 Hz frequency range, with a total excursion of 0.014 inches for 10 minutes each plane.

3.16.5 Shock. Continuous operation when subject to the shock test of W-C-1422, paragraph 4.5.20.5.

3.17 Camera mounting. The camera base plate shall have not less than two standard 1/4-20 threaded holes for mounting the camera. Mounting holes should be 1.5 to 2 inches apart on centers and should be centered on the base mounting plate.

3.18 Dimensions. The camera, less the lens, shall not exceed the following dimensions:

Height	9 inches
Width	8.5 inches
Length	16 inches
Weight	20 pounds

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling, inspection, and test procedures. Sampling, inspection, and test procedures shall be in accordance with W-C-1422.

5. PREPARATION FOR DELIVERY

5.1 Preparations for delivery shall be in accordance with W-C-1422.

6. NOTES

6.1 The notes of W-C-1422 shall apply.

6.2 Ordering data. Purchasers should select the preferred options permitted in W-C-1422 and herein and include the following information in procurement documents:

(a) Specify style, and mode of synchronization and interlace required (see 1.2.1 and 3.5).
(b) When specifying mode 1 or mode 2 synchronization, also specify either EIA RS-170 or EIA RS-330 sync (see 3.9.1 and 3.9.2).

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